Lab 1a - Raw Security Logs to MITRE ATT&CK

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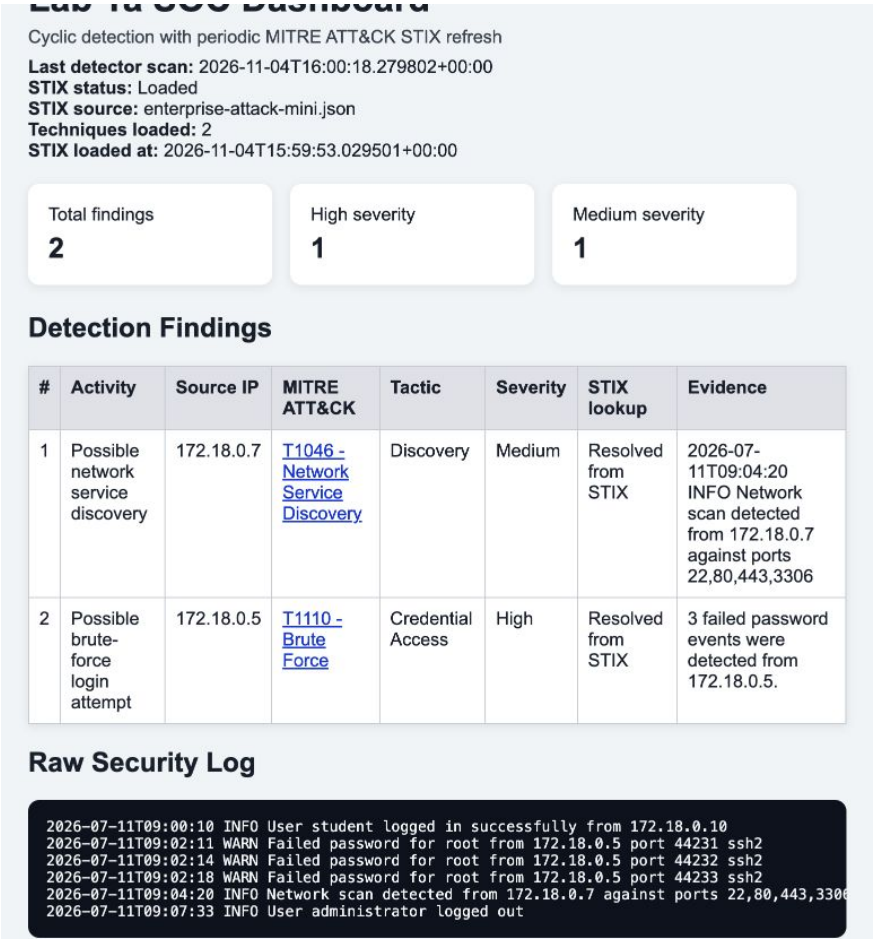
1. Scope and execution

The supplied course project was executed locally with Codex assistance on macOS Apple Silicon, using zsh and Docker Desktop instead of Windows PowerShell. The supplied Python and Docker Compose files were not changed. Only supplied and synthetic logs were used. No network scan or password attack was executed. The optional simulator and full ATT&CK download were not used.

Docker client/server 29.9.2; Docker Desktop 4.97.0. Detector and dashboard were running. Verification records are in evidence/results.zip (docker-version.txt and compose-ps.txt).

2. Baseline and final counts

Baseline: 6 records, 2 findings (1 High, 1 Medium). T1110: 172.18.0.5; T1046: 172.18.0.7. Final C4: 3 findings (1 High, 2 Medium). Both techniques resolved through the local knowledge file. Final evidence is shown with C4 on page 4.



Cyclic detection with periodic MITRE ATT&CK STIX refresh

Last detector scan: 2026-11-04T16:00:18.279802+00:00
STIX status: Loaded
STIX source: enterprise-attack-mini.json
Techniques loaded: 2
STIX loaded at: 2026-11-04T15:59:53.029501+00:00

Total findings	High severity	Medium severity
2	1	1

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible brute-force login attempt	172.18.0.5	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 172.18.0.5.

Raw Security Log

```
2026-07-11T09:00:10 INFO User student logged in successfully from 172.18.0.10
2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2
2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.5 port 44232 ssh2
2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.5 port 44233 ssh2
2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,330
2026-07-11T09:07:33 INFO User administrator logged out
```

Figure 1. Actual baseline dashboard: 2 findings, including 1 High and 1 Medium.

3. Baseline evidence, rules and ATT&CK mapping

Evidence	Rule	Technique	Tactic	Uncertainty
security.log lines 2-4: three failed root passwords from 172.18.0.5 at 09:02:11, 09:02:14 and 09:02:18	At least 3 failures per source IP; High	T1110 Brute Force	Credential Access	Failures do not establish malicious intent or successful login; mistakes or shared source addresses can trigger the same rule.
security.log line 5: scan-labelled record from 172.18.0.7 against 22,80,443,3306	Text contains Network scan detected; Medium	T1046 Network Service Discovery	Discovery	A phrase match does not independently verify packets or distinguish approved scanning.

Full original records: evidence/baseline-security.log. The successful student login at line 1 comes from another IP; it does not prove success by the source that generated the failed-password records.

4. Rule inspection and MITRE enrichment

FAILED_PASSWORD_THRESHOLD defaults to 3; SCAN_INTERVAL_SECONDS defaults to 5. Both are set to those values in Compose. There is no time window. extract_source_ip takes the first token after the literal separator ' from '. create_candidate_findings counts failed-password records per source and emits one T1110 candidate per source above threshold. Each scan-labelled line creates its own T1046 candidate. Repeated identical scan records are not deduplicated. The code lowercases text for phrase matching, but source extraction is literal. The rule severity is assigned by the detector, not supplied by MITRE.

The bundled enterprise-attack-mini.json contains two simplified educational STIX-style objects. Their IDs and version/date fields are teaching fixtures, not authenticated official STIX identifiers or current official metadata. Resolved from STIX means a local lookup succeeded, not that MITRE confirmed an incident. Names, technique links and tactic assignments were checked against the official ATT&CK pages. T1110 is retained at parent-technique level rather than claiming the rule identifies a particular sub-technique.

5. Controlled experiments C1-C4

Experiments were sequential. Every state retains the original scan record. The simulator remained off. After log edits, outputs were collected following automatic rescanning; containers were not restarted for these edits.

C1 - Below threshold

Change: removed baseline line 4, the failure at 09:02:18 from 172.18.0.5.

Prediction: 1 finding; T1110 disappears and original T1046 remains.

Actual: 1 finding; T1046 172.18.0.7. Matched prediction.

Explanation: the original source now has two failed-password events.

Evidence: evidence/c1-security.log and evidence/c1-findings.json.

C2 - Three different sources

Change: replaced the two remaining baseline password failures with:

2026-10-22T10:00:00Z WARN Failed password for test from 192.0.2.51 port 40001 ssh2

2026-10-22T10:00:02Z WARN Failed password for test from 192.0.2.52 port 40002 ssh2

2026-10-22T10:00:04Z WARN Failed password for test from 192.0.2.53 port 40003 ssh2

Prediction: 1 finding; no T1110, original T1046 remains.

Actual: 1 finding; T1046 172.18.0.7. Matched prediction.

Explanation: each password-failure source has count 1, not 3.

Evidence: evidence/c2-security.log and evidence/c2-findings.json.

C3 - Threshold crossing

Change: appended to C2:

2026-10-22T10:01:00Z WARN Failed password for test from 192.0.2.55 port 40001 ssh2

2026-10-22T10:01:02Z WARN Failed password for test from 192.0.2.55 port 40002 ssh2

2026-10-22T10:01:04Z WARN Failed password for test from 192.0.2.55 port 40003 ssh2

Prediction: 2 findings; a new T1110 plus original T1046.

Actual: 2 findings; T1110 192.0.2.55 and T1046 172.18.0.7.

Explanation: the new source reaches the inclusive threshold. Matched prediction.

Evidence: evidence/c3-security.log and evidence/c3-findings.json.

5. Controlled experiments - C4

C4 - Another scan

Change: appended to C3:

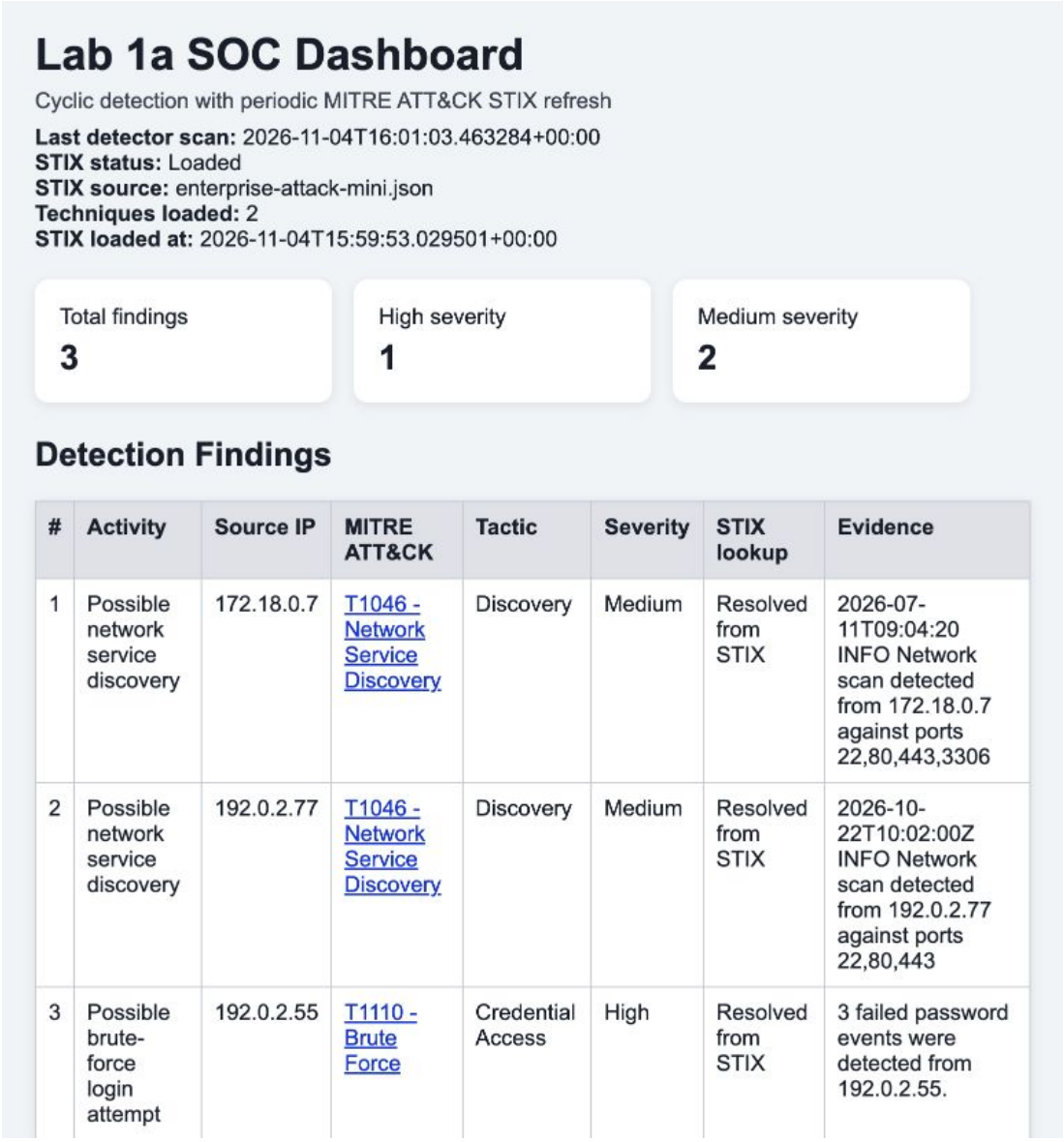
2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443

Prediction: 3 findings; one additional Medium T1046.

Actual: 3 findings; 1 High T1110 and 2 Medium T1046. Matched prediction.

Explanation: the new scan-labelled line matches independently of password counts.

Evidence: evidence/c4-security.log, evidence/c4-findings.json, dashboard-c4.jpg.



Lab 1a SOC Dashboard

Cyclic detection with periodic MITRE ATT&CK STIX refresh

Last detector scan: 2026-11-04T16:01:03.463284+00:00
STIX status: Loaded
STIX source: enterprise-attack-mini.json
Techniques loaded: 2
STIX loaded at: 2026-11-04T15:59:53.029501+00:00

Total findings	High severity	Medium severity
3	1	2

Detection Findings

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2	Possible network service discovery	192.0.2.77	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443
3	Possible brute-force login attempt	192.0.2.55	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 192.0.2.55.

Figure 2. Actual C4 dashboard: 3 findings, including 1 High and 2 Medium. Captured viewport shows the findings; full raw logs are preserved in the evidence archive.

6. SOC narrative

The baseline contains three failed root-password events from one source and one scan-labelled record. The detector groups password failures by source IP and emits a High T1110 candidate when the count reaches three. The scan phrase independently produces a Medium T1046 candidate. Reducing failures to two removed T1110; distributing three failures across three sources also stayed below the per-source threshold. Adding three failures from one new source restored a T1110 candidate, and adding another scan record produced a third finding. These results demonstrate deterministic rules, not confirmed hostile intent or successful compromise. Legitimate mistakes, shared source addresses, or an authorized scanner could explain similar records. Authentication outcomes, approved scanning schedules, firewall flows, and endpoint telemetry would help validate the findings. The detector has no time window and cannot establish attacker identity.

7. Containerization and reflection

Docker provides a reproducible Python environment, separate services and shared log/output mounts; cloud services are not required for these benefits. False positives include user mistakes, approved scanners and aggregation behind one source address. Distributed guessing can be missed because each IP stays below threshold; old failures can accumulate because there is no time window. The scan rule consumes an existing text label rather than detecting packets. Additional authentication, firewall and endpoint records are needed to assess intent, outcome and scope. A finding alone is insufficient to confirm an attack.

8. Cleanup

The clean six-line baseline was restored after saving C4 evidence. The running detector returned to 2 findings; services were then stopped with Compose down. The optional simulator was never started, so no simulator counts are claimed.

9. Sources and supporting evidence

Course project and handout - pinned source snapshot

MITRE ATT&CK; T1110

MITRE ATT&CK; T1046

Official ATT&CK pages checked 4 November 2026. Extract evidence/results.zip into evidence/ to resolve the individual log, JSON and verification paths cited in this report. The TXT file is the editable text companion; screenshots are embedded in this PDF.